ALTER

Product Requirements Document • Current state v1.1.0

Document purpose

This document captures Alter's current product requirements and implementation status. It describes the current-state product, not a future-state roadmap or a claim of validated product-market fit.

Alter – Product Requirements Document (current state, v1.1.0)

1. Product summary

Alter is a personal AI organiser for Android with a FastAPI + MongoDB backend. Users speak or type in English, Hindi or Hinglish. Alter turns what they say into tasks, alarms and grocery items, syncs their Google Calendar and Gmail to-dos into tasks, rings full-screen alarms, and lets them chat with an AI that only answers from their own history.

2. Target users

- Individuals and families who want one place for to-dos, groceries and reminders.
- Mixed-language users (English / Hindi / Hinglish).
- Private-beta testers who install the Android APK.

3. Core requirements

#	Requirement	Status
R1	Voice or typed input: AI decides grocery item vs task vs alarm	Done
R2	Google Calendar sync: today and the next 30 days imported as tasks, every 10 min plus "Sync calendar now"	Done
R3	Chat answers only from the user's own history (tasks, completed tasks, calendar events, memories, past chats); no general knowledge	Done
R4	No beta restrictions or per-user feature limits for signed-in users	Done
R5	Full-screen Android alarms over the lock screen; snooze 5/10/30 min; repeat daily, weekdays, weekly or monthly	Done
R6	Works when AI is unavailable: offline parser for "buy X", "X lena hai", "alarm at 7am" and "alarm in 10 minutes"	Done
R7	Bring-your-own Gemini key (BYOK); app's own Gemini key for everyone else; no Emergent Universal key fallback for users	Done
R8	Feedback / bug reports emailed to	Done

	the admin, with bug counts by app area

4. Features

Home

- One input row: type a task, groceries or an alarm; when the box is empty, the button becomes a mic.
- AI "Your day" summary, today's tasks, grocery and family cards.
- Quick actions: Task with time, Memory, Chat.

Tasks

- Grouped **Today** (including overdue), **Tomorrow**, **Later**; sorted Urgent → High → Medium → Low.
- Priority tag and Alarm tag (with its repeat rule).
- Views: Tasks | History (day navigator) | Groceries.
- "Test my alarm" button.
- "Sync calendar now" button, shown once Google is connected.
- Banner when Android has blocked exact alarms for Alter.
- Task form fields: title, description, date, time, priority, status, assignee, confidential;
- alarm toggle, and a Repeat option (Once / Every day / Weekdays / Every week / Every month) once the alarm is on.

Alarms

- Android: full-screen alarm intent (react-native-notify-kit, SET_ALARM_CLOCK), alarm sound, bypasses Do Not Disturb.
- Alarm tasks ring on time only. Normal timed tasks also get alerts 2 h, 1 h and 5 min before.
- Ringing screen: Mark as done, Snooze 5/10/30 min, Stop.
- Repeating alarms move themselves to the next occurrence; the phone schedules about a week of rings ahead.

Groceries

- Shared with the family, de-duplicated (plural-aware), check off and clear checked.

Chat

- Uses only the user's own data. Every answer lists its sources ("Based on ...").
- Can create and update tasks and memories. Sensitive memories need Save/Dismiss approval.
- Mic button for voice messages.

Memory

- Categories, search, edit and delete.
- Secrets (passwords, OTPs, keys) are blocked from being saved.

Google integration

- "Connect Google account" (gmail.readonly + calendar.readonly).
- Calendar events and Gmail to-dos become tasks; tokens are stored encrypted.

Family

- 6-character join code, task assignment, chore board, shared groceries, family plans (confidential tasks stay private).

Profile and Settings

- Profile: name, description, preferences, response style, voice language, Gemini key card, feedback, sign out, app version.
- Settings: time zone (phone zone, common zones, or typed), AI usage bar, delete account.

Admin (developer account only)

- Analytics: DAU/WAU/MAU, retention, churn, feature use, bugs by area, guardrail metrics.
- AI usage and billing, emergency AI pause, app-wide limits.
- Feedback inbox.
- Budget alert email at 90% of the daily AI cap.

5. Non-functional requirements

- **Privacy:** every query is scoped to the signed-in user; Google tokens and BYOK keys are Fernet-encrypted.
- **Reliability:**
- Automatic Gemini model fallback when one model's quota runs out.
- A 25 s timeout per attempt, after which it switches model.
- Offline parsers for groceries and alarms.
- **Cost:** light models first; only an app-wide cap (3,000 AI calls/day, 50,000/month); 90% budget email.
- **Languages:** English, Hindi, Hinglish for voice, typing and chat.

6. Out of scope / known limits

- The web version can't show lock-screen alarms; alarms ring only while the tab is open.
- The deployed `*.emergent.host` domain serves the backend API only; users use the APK or Expo Go.
- Google OAuth app is in Testing mode, so only listed test users can connect Google.

7. Backlog ideas

- Faster streaming chat replies.
- "What Alter heard" voice confirmation with a fix-it option.
- Warning when Calendar access wasn't granted.
- In-app update prompt.
- Alarm sound choice; list of all alarms.
- Bug-report screenshots; weekly bug digest.